ROUTING AND WAVELENGTH ASSIGNMENT OVER MULTI-FIBER OPTICAL NETWORKS WITH NO WAVELENGTH CONVERSION

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Abstract We consider the routing and wavelength assignment (RWA) optimization problem with no wavelength conversion for multi-fiber WDM networks. We formulate a mixed integer linear program (MILP) that optimizes the three RWA subproblems together, they are: (1) routing of traffic demands; (2) routing of lightpaths; and (3) wavelength assignment for lightpaths. Minimizing network congestion is the aim of our optimization. The RWA problem is computationally intractable for problems of realistic size. We consider several search space dimensionality (SSD) reduction techniques that enhance solvability of the MILP. Our main contribution is a framework for pruning the search space. This framework allows us to compute optimal solutions in a reasonable time that exhibit certain predefined features. Based on the computational complexity of the MILP, we choose in advance, which features of the search space are to be pruned. We have evaluated our framework on an Australian backbone network, and demonstrate that our framework can achieve more than a 50 percent improvement in the minimum level of congestion achievable in comparison to an existing SSD reduction technique reported in the literature.

Keywords: optical networks, MILP, RWA, search space reduction, WDM network design

1. Introduction

1.1 Optical Networks

Wavelength division multiplexing (WDM) and wavelength routing are rapidly becoming the technologies of choice in network infrastructure that must accommodate unprecedented, accelerating demand for bandwidth [Siva Ram Murthy and Gurusamy, 2002]. Underpinning WDM networks are 'single-hop' wavelength-routed all-optical channels called *lightpaths*. Lightpaths allow optical networks to fully capitalize on the high transmission capabilities of WDM because the transmission speed of a lightpath is the same as that of a wavelength in a fiber. A lightpath is set up by using transparent coupling to switch the same colored wavelengths in a sequence of contiguous fibers. Switching of a wavelength from an input fiber to an output fiber is achieved with an optical cross-connect (OXC). The OXC uses an all-optical switching fabric to connect each wavelength arriving at an input port to the corresponding wavelength at a selected output port. Equipping OXCs with wavelength conversion facilities enables switching between different colored wavelengths. However, wavelength conversion increases hardware cost and complexity. The number of input/output ports on an OXC is limited. Typically slow electronic routing is eliminated because the switching operation performed by the OXC takes place in the fast optical domain. Therefore, data packets traversing a lightpath are never delayed by electronic processing. Without lightpaths, data packets must undergo time consuming electro-optic conversion, electronic forwarding and opto-electronic conversion at intermediate nodes.

In a 'single-hop' all-optical network, lightpaths are set up between all end-node (EN) pairs needing bandwidth. A lightpath is limited to the transmission speed of a single wavelength. Multiple lightpaths are set up between those EN pairs needing a transmission speed greater than that offered by a single wavelength. Lightpaths that exist between the same EN pair can be set up on more than one particular sequence of contiguous fibers. In an all-optical network, a *logical link* is a collection of lightpaths that are set up between the same EN pair. Each lightpath in a logical link is assigned a unique wavelength color. The collection of all established logical links is called the *logical topology*. In networks of a practical size, there are often not enough wavelengths to set up the necessary lightpaths to establish an all-optical network (fully-meshed logical topology).

Hence, it may not be possible to directly connect some EN pairs via a single lightpath. These EN pairs communicate through a sequence of contiguous logical links called a *route*. Note that, a route can also consist of one logical link that exists between an EN pair. Data packets carried on lightpaths that are terminated at intermediate core nodes (CNs) are electronically routed to a subsequent lightpath enroute to the destination EN. Consequently, data packets may need to traverse multiple lightpaths (multi-hops), which are routed in the electronic domain. In a multi-hop lightpath network, a logical link may also be set up between two CNs that are each equipped with an electronic router. In this application, there are two competing factors that need to be taken into consideration. First, it is important to carry data packets in the optical domain as far as possible to ensure that delays due to the slow electronic domain are minimized. Second, the transmission speed of a single wavelength is better utilized in multi-hop lightpath networks because a single lightpath can accommodate traffic from several EN pairs, which is known as traffic aggregation.

1.2 Routing and Wavelength Assignment Optimization Problem

Embedding a logical topology over a wavelength routed optical network is an extensively studied and well known optimization problem that is typically formulated as a mixed integer linear program (MILP). This optimization problem consists of the following three subproblems:

1 routing of traffic demands;

2 routing of lightpaths; and

3 wavelength assignment for lightpaths.

All three subproblems are dependent on each other. Therefore, optimizing each subproblem separately is likely to preclude the enforcement of a global optimization criterion [Kuri *et al.*, 2002]. The combination of all three subproblems is called the routing and wavelength assignment (RWA) optimization problem.

A solution to the RWA problem consists of a logical topology (set of logical links) with a wavelength assignment for each lightpath that is established in a logical link. A scheme for routing the traffic demand over the logical topology is also determined. An optimal solution must furthermore optimize a certain network performance measure. The level of network congestion is a common performance measure for the RWA problem. *Congestion* is the maximum amount of traffic carried by any logical link in the logical topology. Minimizing network congestion tends to encourage lightpaths to be set up between EN pairs that carry a heavy traffic load [Krishnaswamy and Sivarajan, 2001].

The RWA problem can be defined as follows. Consider a directed graph $G_p = (V_p, E_p)$. G_p represents the *physical topology* of the network. Each vertex in V_p represents an OXC that may (or may not) include an electronic router. Each edge in E_p represents a multi-fiber transmission link (physical link). The number of wavelengths per fiber and the number of input/output ports on an OXC is a critical constraint in the RWA problem.

Consider another directed graph $G_l = (V_l, E_l)$. G_l represents the *logical topology* of the network. Each vertex in V_l represents an OXC that must include an electronic router. Each edge in E_l represents a *logical link*. The set of vertices V_l is a subset of V_p and can be further partitioned into two mutually exclusive sets of nodes. The first set includes only ENs. Each EN is associated with customers and represents either an origin o or a destination d of a certain route. A traffic demand matrix $\Lambda_{o,d}$ quantifies the amount of traffic flow from origin o to destination d. $\Lambda_{o,d}$ must be zero if there does not exist a sequence of contiguous logical links in V_l that define a route from origin o to destination d. The remaining nodes in V_l are CNs. No route originates or terminates at a CN. Those CNs in V_p that are not in V_l do not include an electronic router.

Solving the RWA problem involves finding a graph G_l in conjunction with a routing and wavelength assignment for each lightpath set up in E_l. A scheme for routing the traffic demand Λ over G_l is also determined.

The RWA problem is intractable for networks of realistic size [Bienstock and Gueluek, 1995].

1.3 Previous Contributions

In [Kuri *et al.*, 2002], [Peuch *et al.*, 2002], [Cinkler *et al.*, 2000], [Leonardi *et al.*, 2000], [Mukherjee *et al.*, 1996]and [Ramaswami and Sivarajan, 1996]the RWA problem is formulated as a MILP. However, in each formulation, either the wavelength assignment subproblem (see Subsection 1.2) is neglected or wavelength conversion is assumed. Wavelength conversion may increase hardware costs. In [Banerjee and Mukherjee, 2000], the RWA problem with no wavelength conversion is formulated as a non-linear integer program. Solving a non-linear program is intractable for large problems. Wavelength conversion is assumed by removing the non-linear constraint.

In [Harai *et al.*, 2000], the RWA problem is formulated as a MILP. A set of routes between each EN pair and a wavelength assignment for logical links is predetermined and provided as an input to the MILP. Solving the MILP provides a routing scheme for the traffic demand that maximizes network throughput. The solution is only optimal for the set of routes and the wavelength assignment that is predetermined.

In [Krishnaswamy and Sivarajan, 2001], the RWA problem with no wavelength conversion is formulated as a MILP. However, each link is limited to a single fiber. Multiple fibers on each link enable lightpaths, which are assigned a common wavelength color, to exist between the same EN pair.

Heuristic algorithms are typically employed to cope with intractable instances of the RWA problem. In [Krishnaswamy and Sivarajan, 2001], a two step heuristic is proposed. The first step involves a well known maximum rounding algorithm to configure a logical topology with a tentative wavelength assignment that may contain wavelength clashes. The second step employs a vertex coloring algorithm to establish a clash free wavelength assignment for the logical topology configured in the first step. A shortcoming of the two step heuristic is that the second step may require additional wavelengths to establish a clash free wavelength assignment.

The objective of the MILP formulated in [Banerjee and Mukherjee, 2000]is to minimize the average hop length of logical links. It is shown that solutions within 0.1 percent of the lower bound provided by the LP-relaxation of the MILP are achievable by terminating the branch-and-bound algorithm after only a few iterations. This allows for the computation of near-optimal solutions in a reasonable time.

In [Kuri *et al.*, 2002]a size reduction technique is proposed to cope with large instances of the RWA problem. The search space of the MILP is pruned by considering only the shortest routes between each EN pair.

1.4 Our Contribution

We consider the RWA optimization problem with no wavelength conversion. We resolve the shortcomings of the formulations proposed by [Krishnaswamy and Sivarajan, 2001]and [Harai *et al.*, 2000]. We formulate a MILP that optimizes the three RWA subproblems together (see Subsection 1.2). Our optimization objective is to minimize network congestion. We do not predetermine a wavelength assignment for logical links as in [Harai *et al.*, 2000]. Our MILP is not limited to a single fiber on each physical link as in [Krishnaswamy and Sivarajan, 2001].

To cope with intractable instances of the RWA problem, we propose an alternative approach that does not employ a heuristic algorithm. Our approach is motivated by the search space dimensionality (SSD) reduction techniques proposed in [Zalesky *et al.*, (in press)]and [Kuri *et al.*, 2002].

We propose to reduce the SSD of the MILP to ensure that a solution can be computed in a reasonable amount of time. The SSD is reduced by pruning impractical solutions from the search space. For example, we can prune solutions that are likely to violate restrictions associated with network routing protocols.

We propose a framework for pruning the search space. Our framework allows us to compute solutions that exhibit predefined features such as no cyclic routes (see Subsection 3.1), only shortest logical links (see Subsection 3.2), only shortest routes (see Subsection 3.3) and only routes that do not exceed certain hop-limits (3.4). Based on the computational complexity of the MILP, we decide in advance, which features of the search space need pruning to ensure a solution can be computed in a reasonable amount of time. Having the ability to prune only certain (undesirable) features allows us, in many cases, to compute better solutions than considering only one SSD reduction technique as in [Kuri *et al.*, 2002].

The remainder of this paper is organized as follows. In Section 2, we precisely formulate the multi-fiber RWA problem with no wavelength conversion as a MILP. In Section 3, we describe the SSD reduction techniques used in our framework. In Section 4, we consider an illustrative example. We compute the minimum level of network congestion achievable in a four node network when certain features of the search space are pruned. In Section 5, we apply our framework to an Australian backbone network (ABNET). We demonstrate that in some cases our framework shows more than a 50 percent improvement in the minimum level of network congestion achievable in comparison to the SSD reduction technique proposed in [Kuri *et al.*, 2002].

2. Problem Formulation

In this section the multi-fiber RWA optimization problem with no wavelength conversion is formulated as a MILP.

2.1 Notation

We introduce the following notation for indexing variables that describe the network:

m	EN pair needing bandwidth;
n	node in the network;
j	physical link in the physical topology;
f	fiber on a physical link;
w	wavelength on a fiber;
l	logical link in the logical topology;
i	lightpath in a logical link;
p	route between EN pair.

2.2 Given Parameters

We assume the following sets are known a priori:

$\mathcal{M}$	EN pairs in the network;
$\mathcal{N}$	nodes in the network;
$\mathcal{J}$	physical links in the network;
$\mathcal{F}$	fibers on a physical link;
$\mathcal{W}$	wavelengths on a fiber;
$\mathcal{L}$	all conceivable logical links in the logical topology;
$\mathcal{P}_m$	all conceivable routes between EN pair $m \in \mathcal{M}$;
$\mathcal{H}_m^l$	($\mathcal{H}_m^l \subset \mathcal{P}_m$) routes between EN pair $m \in \mathcal{M}$ that traverse logical link $l \in \mathcal{L}$;
$\mathcal{K}_l$	($\mathcal{K}_l \subset \mathcal{J}$) physical links traversed by logical link $l \in \mathcal{L}$;
$\mathcal{O}_j$	($\mathcal{O}_j \subset \mathcal{L}$) logical links traversed by physical link $j \in \mathcal{J}$;
$\mathcal{D}_n^{\text{in}}$	($\mathcal{D}_n^{\text{in}} \subset \mathcal{L}$) logical links incident to node n;
$\mathcal{D}_n^{\text{out}}$	($\mathcal{D}_n^{\text{out}} \subset \mathcal{L}$) logical links incident from node n.

The set $\mathcal{P}_m$ is constructed from $\mathcal{L}$. Each route in $\mathcal{P}_m$ consists of a sequence of contiguous logical links chosen from $\mathcal{L}$. The set $\mathcal{P}_m$ contains every conceivable route that can be constructed from $\mathcal{L}$.

Solving the MILP determines an optimal subset of routes $\overline{\mathcal{P}}_m \subseteq \mathcal{P}_m$ and a corresponding subset of logical links $\overline{\mathcal{L}} \subseteq \mathcal{L}$. Lightpaths are configured for all $l \in \overline{\mathcal{L}}$ so that each logical link l contains a sufficient number of lightpaths (enough capacity) to carry the traffic that is routed over it.

Furthermore, we assume the following parameters are known:

Δ	maximum degree of the logical topology; i.e., the maximum number of input/output ports on an OXC;
C	maximum capacity per wavelength;
Π	maximum number of lightpaths configurable in a logical link;
Λ_m	average traffic demand between EN pair $m \in \mathcal{M}$.

Note that $\Pi = |\mathcal{W}| \times |\mathcal{F}|$.

2.3 Variables

The MILP is formulated with the following integer, binary and fractional variables:

$\delta^i_{w,l}$	$\delta^i_{w,l} = 1$ if lightpath i in logical link l is assigned to wavelength w, otherwise $\delta^i_{w,l} = 0$;
$\delta^i_{j,w,l}$	$\delta^i_{j,w,l} = 1$ if lightpath i in logical link l traversing physical link j is assigned to wavelength w, otherwise $\delta^i_{j,w,l} = 0$;
$\delta^f_{j,w,l}$	$\delta^f_{j,w,l} = 1$ if logical link l is assigned to wavelength w on fiber f of physical link j, otherwise $\delta^f_{j,w,l} = 0$;
π_l	$\pi_l \in \{0, \ldots, \Pi_l\}$ is the number of 'active' lightpaths in logical link l assigned to a wavelength;
$\lambda_{m,p}$	$0 \leq \lambda_{m,p} \leq \Lambda_m$ is the amount of traffic between EN pair m aggregated to path $p \in \mathcal{M}_m$;
λ	λ is maximum amount of traffic carried by any logical link in $\mathcal{L}$. λ quantifies network congestion.

The δ-group of binary variables specify a subset of logical links $\overline{\mathcal{L}} \subseteq \mathcal{L}$ that correspond to an optimal logical topology for the problem. They also specify a wavelength assignment for each 'active' lightpath in a logical link. The integer variable π_l enumerates the total number of 'active' lightpaths in logical link l. The λ-group of fractional variables specify a subset of routes $\overline{\mathcal{P}}_m \subseteq \mathcal{P}_m$ that correspond to an optimal routing scheme for the traffic demand. They also specify the portion of traffic between EN pair m aggregated to each route in $\overline{\mathcal{P}}_m$.

2.4 Objective

$$\min \lambda$$

This objective seeks to minimize network congestion.

2.5 Constraints

Traffic Routing Constraints.

$$\sum_{p \in \mathcal{P}_m} \lambda_{m,p} = \Lambda_m \quad \forall m \in \mathcal{M} \tag{1}$$

$$\sum_{m \in \mathcal{M}} \sum_{p \in \mathcal{H}^l_m} \lambda_{m,p} \leq C \times \pi_l \quad \forall l \in \mathcal{L} \tag{2}$$

$$\sum_{m \in \mathcal{M}} \sum_{p \in \mathcal{H}^l_m} \lambda_{m,p} \leq \lambda \quad \forall l \in \mathcal{L} \tag{3}$$

(1) forces traffic between EN pair m to be partitioned across a subset of the routes in $\mathcal{P}_m$. (2) ensures that the amount of traffic carried by any logical link does not exceed its capacity. (3) quantifies network congestion.

Logical Topology Degree Constraints.

$$\sum_{l \in \mathcal{D}_n^{\text{in}}} \pi_l \leq \Delta \quad \forall n \in \mathcal{N} \tag{4}$$

$$\sum_{l \in \mathcal{D}_n^{\text{out}}} \pi_l \leq \Delta \quad \forall n \in \mathcal{N} \tag{5}$$

(4) ensures that the number of lightpaths incident to node n does not exceed Δ. (5) ensures that the number of lightpaths incident from node n does not exceed Δ. For a Δ-regular logical topology the inequalities in (4) and (5) are replaced with equalities.

Wavelength Constraints.

$$\sum_{w \in \mathcal{W}} \delta_{w,l}^{i} \leq 1 \quad \forall l \in \mathcal{L}, i \in \{1, \ldots \Pi\} \tag{6}$$

$$\sum_{i=1}^{\Pi} \sum_{w \in \mathcal{W}} \delta_{w,l}^{i} = \pi_l \quad \forall l \in \mathcal{L} \tag{7}$$

(6) ensures no more than one wavelength is assigned to a lightpath. (7) enumerates the number of 'active' lightpaths in a logical link.

Fiber Constraints.

$$\sum_{l \in \mathcal{O}_j} \delta_{j,w,l}^{f} \leq 1 \quad \forall j \in \mathcal{J}, f \in \mathcal{F}, w \in \mathcal{W} \tag{8}$$

$$\sum_{l \in \mathcal{O}_j} \sum_{i=1}^{\Pi} \delta_{j,w,l}^{i} \leq |\mathcal{F}| \quad \forall j \in \mathcal{J}, w \in \mathcal{W} \tag{9}$$

$$\sum_{f \in \mathcal{F}} \delta_{j,w,l}^{f} = \sum_{i=1}^{\Pi} \delta_{j,w,l}^{i} \quad \forall l \in \mathcal{L}, j \in \mathcal{K}_l, w \in \mathcal{W} \tag{10}$$

(8) ensures no more than one logical link can be assigned to a specific wavelength on a fiber. (9) ensures that the number of lightpaths in a physical link that are assigned the same wavelength can not exceed the number of fibers in that physical link. (9) ensures that each 'active' lightpath is assigned to only one fiber.

Continuity of Logical Link Constraint.

$$\sum_{j \in \mathcal{K}_l} \delta_{j,w,l}^{i} \leq |\mathcal{K}_l| \delta_{w,l}^{i} \quad \forall l \in \mathcal{L}, i \in \{1, \ldots, \Pi\}, w \in \mathcal{W} \tag{11}$$

(11) forces a logical link to be established if it contains an 'active' lightpath.

3. Search Space Dimensionality Reduction Techniques

State-of-the-art optimization packages such as ILOG CPLEX cannot compute an optimal solution for the MILP we formulate in a reasonable amount of time. Multi-commodity flow optimization problems, such as the RWA problem we consider, are known to be NP-hard, see [Hockenbaum, 1997]). Note that, an optimization problem is NP-hard if there is no polynomial time algorithm that optimally solves the problem for all input cases.[1] In the worst case, optimally solving the MILP we formulate is possible only by exhaustive enumeration.

Heuristic algorithms are typically employed to generate approximate solutions for intractable instances of the RWA problem, see [Cinkler *et al.*, 2000]and [Leonardi *et al.*, 2000]for a survey. Near-optimal solutions are generated by the heuristic proposed in [Krishnaswamy and Sivarajan, 2001]. However, heuristic solutions may consist of routing schemes that are not practical. For example, routing schemes that send the same data packets across the same fiber more than once are likely to violate restrictions imposed by network routing protocols.

We propose to cope with NP-hardness by reducing the SSD of the MILP. In particular, we propose to reduce the dimensionality of the variable space by disallowing certain undesirable routes that we call cyclic. The variable space of the MILP we formulate grows as

$$O\left(|\cup_{m\in\mathcal{M}}\mathcal{P}_m| + |\mathcal{J}||\mathcal{W}||\mathcal{L}|(\Pi + |\mathcal{F}|)\right).$$

For most instances, the term $O\left(|\cup_{m\in\mathcal{M}}\mathcal{P}_m|\right)$ will dominate.

3.1 No Cyclic Routes

A route between an EN pair consists of a sequence of contiguous logical links. We call any such route *cyclic* if it utilizes more than one input port or more than one output port on any particular OXC. For example, in Fig. 1, traffic from node 1 to node 3 is offered two possible routes. The route consisting of the single logical link $1 \to 2 \to 3$ is non-cyclic (*acyclic*). However, the alternative route consisting of the sequence of contiguous logical links $1 \to 2$ and $2 \to 1 \to 3$ is cyclic because it utilizes two output ports on the OXC belonging to node 1. The cyclic route is wasteful because traffic travels back and forth between node 1 and 2; hence the name cyclic.

For large instances of the RWA problem, disallowing cyclic routes may not reduce the SSD enough to allow for a reasonable computation

[1] Assuming that $P \neq NP$.

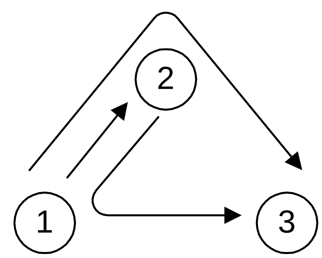

Figure 1. Example of cyclic and non-cyclic routes. Each edge represents a logical link.

time. In such instances, we propose to reduce the SSD even further by disallowing additional routes. In particular, we only consider those routes between each EN pair that consist of the 'shortest' logical links. If the SSD is still to large, we achieve even further reduction by considering only the 'shortest' routes between each EN pair as in [Kuri *et al.*, 2002].

3.2 Shortest Logical Links

We define the set of *shortest logical links* from vertex $o \in V_p$ to vertex $d \in V_p$ to consist of those logical links that traverse the minimum number of edges in E_p. SSD reduction is achieved by considering only those routes between each EN pair that use the shortest logical links.

3.3 Shortest Routes

We define the set of *shortest routes* between each EN pair to consist of those routes that traverse the minimum number edges between that EN pair. SSD reduction is achieved by considering only the shortest routes between each EN pair. Considering only the k-shortest routes between each EN pair is a variation of this SSD reduction technique.

Although this technique is proposed in [Kuri *et al.*, 2002], it is not well suited to the MILP formulated in this paper. In particular, disallowing routes that are not the shortest may only be possible by including additional constraints that explicitly disallow such routes. Additional constraints increase the SSD and cancel out the reduction achieved foremost. This shortcoming is avoided in the MILP we formulate because a unique variable $\lambda_{m,p}$ is enumerated for every route. Disallowing a route that is not the shortest is easily achieved by not enumerating the corresponding $\lambda_{m,p}$ variable.

In Fig. 2 we consider a three node topology in which a physical link exists between each node. We show the routes enumerated from node 1 to node 3 for each SSD reduction technique discussed so far. Route

(a) consists of one logical link that is not a shortest logical link. Route (b) consists of two shortest logical links. Route (c) is the shortest route from node 1 to node 3.

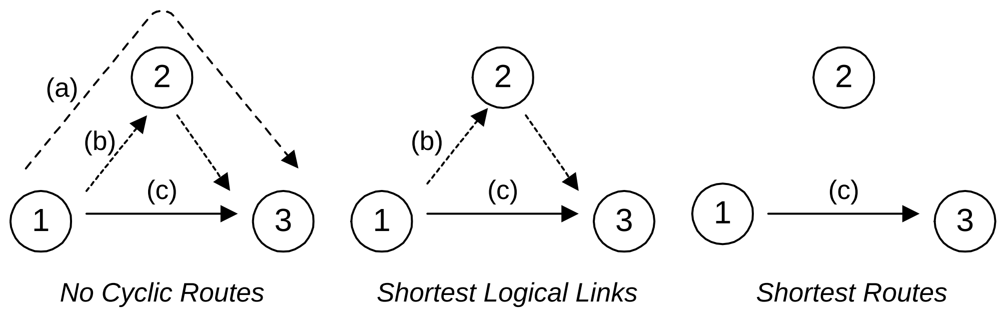

Figure 2. Routes enumerated from node 1 to node 3 for three SSD reduction techniques.

3.4 Hop-Limits

We define the *logical hop-length* of a route as the total number of logical links traversed by that route. We define the *physical hop-length* of a logical link as the total number of physical links (fibers) traversed by that logical link. For example, in Fig. 2, route (a) and (c) are of logical length 1 and route (b) is of logical length 2. Route (a) consists of a single logical link that is of physical length 2. Route (b) consists of two logical links, which are each of physical length 1. Route (c) consists of a single logical link, which is of physical length 1. The SSD can be reduced by imposing a logical and physical hop limit.

3.5 The Framework

We propose a framework for pruning the search space of the MILP. Our framework prescribes an appropriate SSD reduction technique based on the following three criteria:

1 SSD of the MILP;

2 amount of time available to compute a solution; and

3 predefined features the solution must exhibit.

We use the total number of enumerated routes $|\cup_{m\in\mathcal{M}}\mathcal{P}_m|$ to roughly measure the SSD of the MILP. We call the MILP *solvable* if the number of routes enumerated is below a certain threshold. This threshold is chosen to ensure that a solution can be computed in a reasonable amount of time. (In the many cases we have solved, we have observed that the time required to compute a solution is roughly proportional to the number of routes enumerated.) Applying an SSD reduction technique reduces the total number of routes enumerated. In Fig. 3, we show the variable space corresponding to some of our SSD reduction techniques.

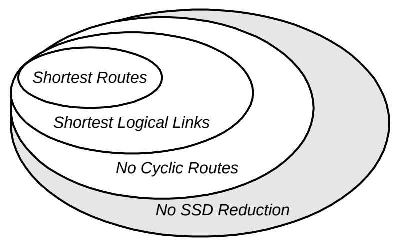

Figure 3. Portions of the variable space corresponding to each SSD reduction technique. Not to any scale.

Our framework is based on the following steps:

1 Prune all solutions that satisfy undesirable predefined features that the final solution must not exhibit. For example, network routing protocols may not allow cyclic routes. In this case, cyclic routes are a predefined feature that is removed from the search space.

2 Measure the SSD and determine if the MILP is solvable; that is, determine if the number of routes enumerated is below the threshold corresponding to the time available to compute a solution. If so, solve the MILP and stop, otherwise consider (3).

3 Sequentially consider each SSD reduction technique in the following order:

   (a) no cyclic routes;
   (b) shortest logical links; and
   (c) shortest routes.

   Impose a logical and physical hop-limit. Repeat (2).

4 The amount of time available to compute a solution is not realistic if the MILP is still not solvable when only shortest path routes are enumerated.

In the next section, we compare the minimum level of network congestion achievable in a small network by applying the SSD reduction techniques discussed in this section. We also compute the congestion achievable when the MILP is solved exactly.

4. Illustrative Example

The RWA problem considered in this section is solved for the four node topology shown in Fig 4.

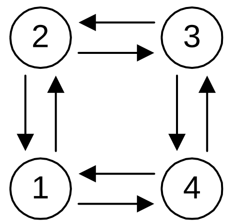

Figure 4. Four node physical topology. Each edge represent a physical link (fiber).

We consider the traffic demand matrix Λ_m shown in Table 1.

Table 1. Traffic demand matrix Λ_m. Each entry is computed at random from a uniform distribution on $[0, 1]$.

0	0.4565	0.9218	0.4103
0.2311	0	0.7382	0.8936
0.6068	0.8214	0	0.0579
0.4860	0.4447	0.4057	0

Setting up the MILP involves enumerating the set of conceivable logical links $\mathcal{L}$. There is a total of 24 conceivable logical links. The set of conceivable routes $\mathcal{P}_m$ is constructed next by enumerating every possible sequence of contiguous logical links that define a route between EN pair $m \in \mathcal{M}$. When we apply one of the SSD reduction techniques, we only enumerate some of the routes between each EN pair.

Table 2 shows the total number of routes enumerated by applying each of the SSD reduction techniques. The SSD is reduced by enumerating fewer routes.

Table 2. Total number of routes $|\cup_{m \in \mathcal{M}} \mathcal{P}_m|$ enumerated in the four node network shown in Fig. 4.

No SSD Reduction	*No Cyclic Routes*	*Shortest Logical Links*	*Shortest Routes*
312	56	48	24

In this section, we limit each physical link to one fiber; that is, $|\mathcal{F}| = 1$. A multi-fiber network is considered in Section 5. We also set $C = \infty$, which is equivalent to relaxing the wavelength capacity constraint (2).

In Table 3, we compare the minimum level of congestion achievable for each SSD reduction technique we consider. Since the network topology considered is small, we can also solve the MILP with the entire search space. We determine the minimum level of congestion achievable by varying the degree of the logical topology Δ and the number of wavelengths per fiber $|\mathcal{W}|$.

Table 3. Minimum level of congestion achievable in the four node network shown in Fig. 4.

Δ	$\|\mathcal{W}\|$	*No SSD Reduction*	*No Cyclic Routes*	*Shortest Logical Links*	*Shortest Routes*
1	1	3.2529	3.4783	3.4783	Infeasible
2	1	1.4820	1.4820	1.4820	1.4820
2	2	1.1144	1.1247	1.1247	1.4820
3	1	1.4820	1.4820	1.4820	1.4820
3	2	0.7410	0.7410	0.7410	0.8214
3	3	0.6957	0.6957	0.6957	0.8214
4	1	1.4820	1.4820	1.4820	1.4820
4	2	0.7410	0.7410	0.7410	0.8214
4	3	0.5164	0.5164	0.5164	0.8214
4	4	0.5164	0.5164	0.5164	0.8214

In Table 3, the Δ column denotes the degree of the logical topology. This corresponds to the maximum number of input/output ports on an OXC. The $|\mathcal{W}|$ column denotes the number of wavelengths available on each fiber.

The purpose of Table 3 is to quantify the increase in the minimum level of network congestion when certain routes are disallowed. Disallowing cyclic routes increases the minimum level of congestion by no more than seven per cent for the instances simulated in Table 3. Considering only shortest logical links and shortest routes increases congestion by no more than 36 percent. For the case when $\Delta = 1$ and $|\mathcal{W}| = 1$, the MILP is infeasible if only shortest path routes are considered.

5. Australian Backbone Network

In this section, we solve the RWA problem for an Australian backbone network (ABNET) according to our framework. We believe that the network topology shown in Fig. 5 is an accurate representation of the ABNET.

The traffic demand matrix shown in Fig. 4 is proportional the population of each city in the ABNET.

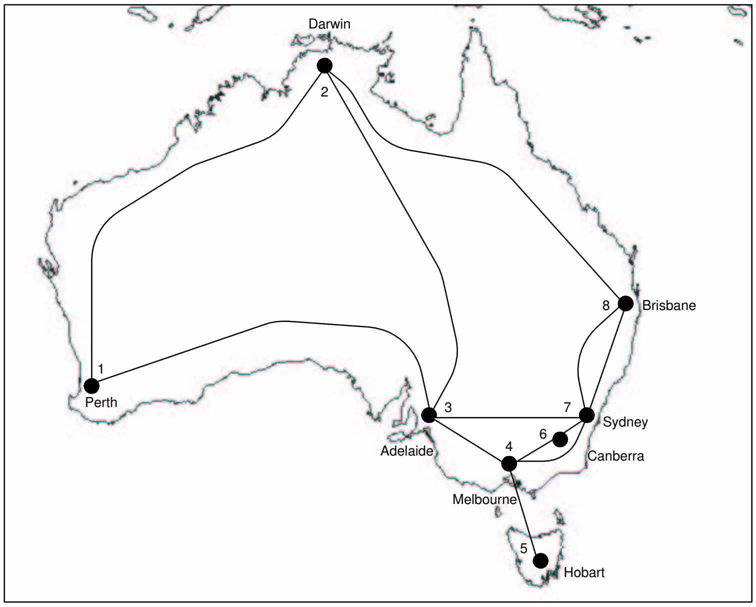

Figure 5. Australian Backbone Network (ABNET).

Table 4. Traffic demand matrix Λ_m for the ABNET.

0	0.568	0.956	1.870	0.608	0.653	2.109	1.160
0.568	0	0.458	1.372	0.110	0.155	1.611	0.662
0.956	0.458	0	1.760	0.498	0.543	1.999	1.051
1.870	1.372	1.760	0	1.412	1.457	2.913	1.965
0.608	0.110	0.498	1.412	0	0.195	1.651	0.703
0.653	0.155	0.543	1.457	0.195	0	1.696	0.747
2.109	1.611	1.999	2.913	1.651	1.696	0	2.204
1.160	0.662	1.051	1.965	0.703	0.747	2.204	0

In this section, we relax the logical topology degree constraints (3) and (4). We assume the capacity of wavelength is 3; that is, we set $C = 3$.

The RWA problem for the ABNET is intractable. We use our framework (see Subsection 3.5) to compute a solution in a reasonable amount of time. In Table 5, we show the total number of routes enumerated by applying each of the SSD reduction techniques. The P and L columns correspond to physical and logical hop-limits, respectively. Table 5 quantifies the SSD of the MILP for each of the reduction techniques with hop-limits considered.

Table 5. Total number of routes $|\cup_{m\in\mathcal{M}}\mathcal{P}_m|$ enumerated in ABNET

P	L	*No SSD Reduction*	*No Cyclic Routes*	*Shortest Logical Links*	*Shortest Routes*
1	3	**184**	**184**	**184**	**70**
2	2	804	**506**	**306**	**116**
3	2	3904	1424	**390**	**130**
2	3	5968	1770	948	**130**
3	3	65058	4116	1164	**144**

To ensure a reasonable computation time, we only solve the RWA problem for those SSD reduction techniques that do not exceed a maximum threshold of 600 routes. Techniques satisfying this threshold are shown with bold face in Table 5. In Table 6 we show the minimum level of congestion achievable for each technique shown with bold face in Table 5.

Table 6. Minimum level of congestion in ABNET

P	L	$\|\mathcal{W}\|$	$\|\mathcal{F}\|$	*No SSD Reduction*	*No Cyclic Routes*	*Shortest Logical Links*	*Shortest Routes*
1	3	1	1	Infs.	Infs.	Infs.	Infs.
1	3	2	1	5.3870	5.3870	5.3870	Infs.
2	2	1	1		Infs.	Infs.	Infs.
2	2	2	1		2.6935	2.6935	Infs.
2	2	3	1		1.8443	2.0887	2.9130
2	2	4	1		1.5135	1.7903	2.9130
2	2	1	4		1.5135	1.7903	2.9130
2	2	5	1		1.4120	1.7903	2.9130
3	2	1	1			Infs.	Infs.
3	2	2	1			2.6935	2.9130
3	2	3	1			2.1276	2.9130
3	2	4	1			1.7903	2.9130
3	2	2	2			1.7903	2.9130
3	2	5	1			1.7902	2.9130
3	2	1	5			1.7902	2.9130
2	3	1	1				Infs.
2	3	2	1				Infs.
2	3	3	1				2.9130
2	3	1	3				2.9130
3	3	1	1				Infs.
3	3	2	1				Infs.
3	3	3	1				2.9130
3	3	1	3				2.9130

In Table 6, the P and L columns correspond to physical and logical hop-limits, respectively. The $|\mathcal{W}|$ column corresponds to the number of wavelengths available on each fiber. The $|\mathcal{F}|$ column corresponds to the number of fibers available on each physical link. The RWA problem is infeasible for entries in Table 6 marked with 'Infs.'.

As shown in Table 6, our framework offers more than a 50 percent improvement on the minimum level of congestion in comparison to only considering the shortest route SSD reduction technique as in [Kuri *et al.*, 2002].

In Table 6, for the all the cases simulated, replacing all k-wavelength fibers with k one-wavelength fibers does not reduce the minimum achievable level of network congestion. This suggests that wavelength conversion offers no benefit in these cases.

All the results presented here are computed with the CPLEX optimization package with default parameter settings. Solving times do not exceed 10 minutes except when the MILP is infeasible. A longer time is usually required to determine infeasibility because the entire branch-and-bound tree requires exploration.

6. Conclusion

In this paper we have considered the RWA optimization problem with no wavelength conversion for multi-fiber WDM-networks. We have formulated a MILP that optimizes all three subproblems of the RWA problem together. The aim of our optimization is to minimize network congestion.

Whereas most authors employ a heuristic algorithm to cope the NP-hardness of the RWA problem, we have opted to consider several SSD reduction techniques. For a four node network, we compared the minimum level of network congestion achievable for each SSD technique we consider. We demonstrated that techniques such as disallowing cyclic routes can drastically reduce the SSD of the MILP and only increase the minimum level of congestion by a small amount.

Our main contribution is a framework for pruning the search space. Our framework allows us to compute solutions that exhibit predefined features. Based on the computational complexity of the MILP and the amount of time available to compute a solution, our framework prescribes a certain reduction technique to implement. We are therefore able to strike a balance between the minimum level of congestion achievable and the time available to compute a solution. For the ABNET, we demonstrated that our framework can achieve more than a 50 percent improvement in the minimum level of congestion achievable in comparison to an existing SSD reduction technique reported in the literature.

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